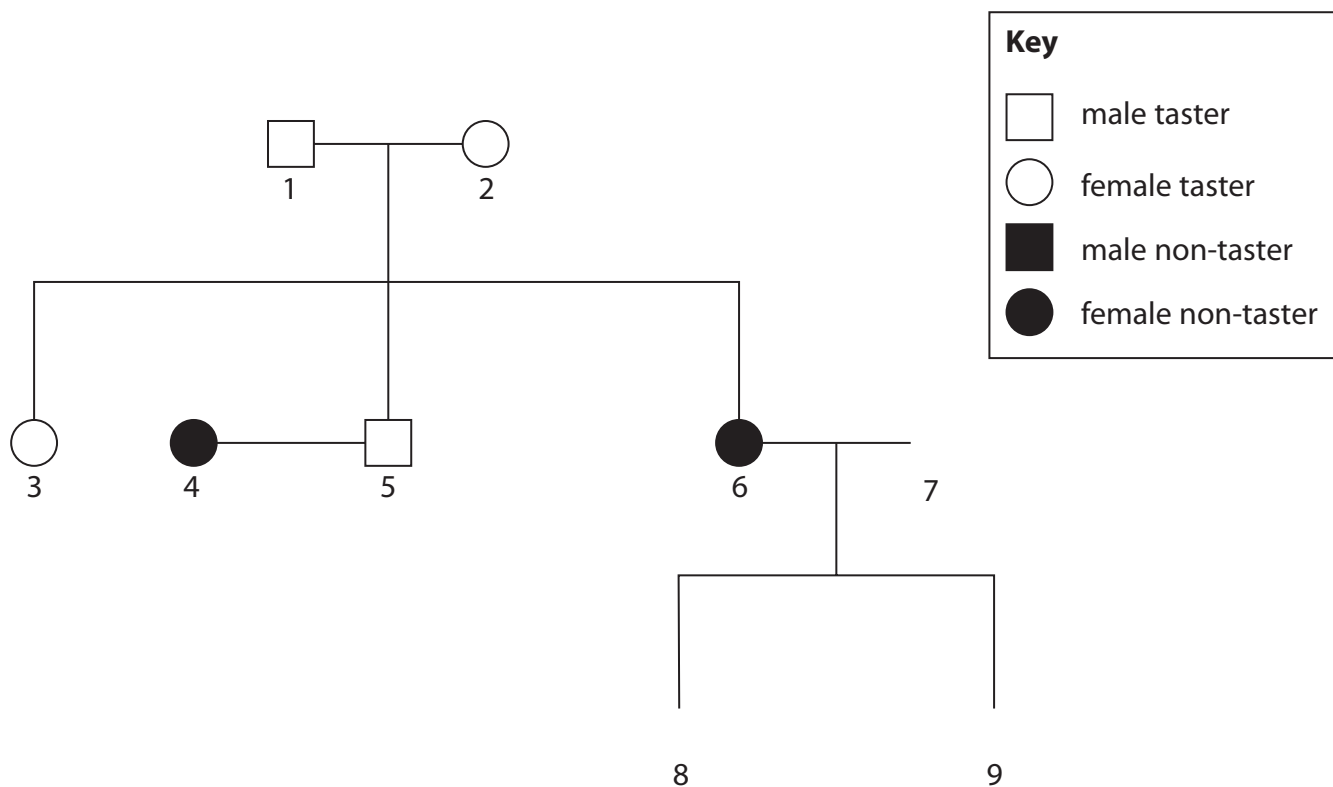


- 4 Phenylthiocarbamide (PTC) is a chemical that has a very bitter taste to some individuals (tasters).

The ability to taste PTC is determined by a gene that codes for a bitter-taste receptor on the tongue.

The pedigree diagram shows some of the tasters and non-tasters in a family.



(a) Complete the diagram to show the following information:

(1)

- individual 7 as a male taster
- individual 8 as a male non-taster
- individual 9 as a female taster.



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(b) Describe the difference between each of the following pairs of terms, using the information in the pedigree diagram to illustrate your answer.

(i) Gene and allele

(2)

(ii) Genotype and phenotype

(2)

(c) Explain which is the dominant allele.

Use the information in the pedigree diagram to support your answer.

(2)



P 6 5 8 1 2 A 0 1 1 3 2

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(d) Explain why this gene is unlikely to be located on the X chromosome.

Use the information in the pedigree diagram to support your answer.

(2)

(Total for Question 4 = 9 marks)

